Cantera vs NN | Test IC 78 | NRMSE (util.py)

T0 = 962.3 K P0 = 2.443 atm phi0 = 1.140 | Repaired self-fed, 10 ms

$$\mathrm{NRMSE} = \sqrt{\mathrm{mean}[(y_{Cantera} - y_{NN})^2]} / \sqrt{\mathrm{mean}[y_{Cantera}^2]}$$

Model / same-state mean	Training	T	Y_{H_2}	Y_H	Y_O	Y_{O_2}	Y_{OH}	Y_{H_2O}	Y_{HO_2}	$Y_{H_2O_2}$	Y_{N_2}	10 ms status
1 ANN10	**10k**	4.72e-05	2.63e-05	3.24	4.29e+03	6.52e-05	1	1	1.07	2.06	0	Complete
2 P-only DeepONet10	**10k**	3.54e-05	1.95e-05	1	0.876	4.00e-05	0.999	0.869	0.848	0.718	0	Complete
2 P-only DeepONet10	**long**	0.91	0.519	7.83e+04	5.98e+07	0.743	2.52e+08	3.49e+04	5.36	48.8	0	Complete
3 ANN10 + P-only DeepONet10	**10k**	0.16	0.0496	8.47e+05	6.63e+07	0.166	1.05e+08	1.77e+03	16.5	4.06	0	Complete
3 ANN10 + P-only DeepONet10	**long**	0.516	0.234	4.10e+04	7.65e+06	0.288	3.86e+07	1.73e+04	11.6	103	0	Complete
4 T/P DeepONet10	**10k**	—	—	—	—	—	—	—	—	—	—	**Stop 1.084 ms**
4 T/P DeepONet10	**long**	4.54e-05	2.58e-05	0.982	0.991	5.41e-05	0.992	0.998	0.992	0.999	0	Complete
5 ANN10 + T/P DeepONet10	**10k**	1.86	0.777	1.64e+06	1.13e+06	0.886	3.31e+07	5.98e+04	8.67	37.4	0	Complete
5 ANN10 + T/P DeepONet10	**long**	6.30e-04	2.92e-04	22.3	11	4.71e-04	17.9	15.1	2.83	10.3	0	Complete
6 T/P/ϕ DeepONet10	**10k**	4.54e-05	2.59e-05	1	0.884	5.42e-05	0.997	1	1	1	0	Complete
6 T/P/ϕ DeepONet10	**long**	5.35e-05	2.92e-05	2.59	1.21	5.90e-05	1.9	1.23	1.02	1.06	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**10k**	4.54e-05	2.58e-05	1	0.819	5.42e-05	0.995	0.999	0.999	1	0	Complete
7 ANN10 + T/P/ϕ DeepONet10	**long**	0.473	0.575	1.67e+07	2.70e+06	0.642	3.16e+08	3.03e+04	8.6	36.9	0	Complete

Exact util.py defaults: norm="rms", reference A=Cantera. Every time sample has equal weight; entries are dimensionless ratios, without x100.
N/A: zero Cantera RMS with a nonzero prediction. Constant nonzero references, including N2, are valid. —: prediction stopped early; full-horizon NRMSE is unavailable.
ANN stays at 10k in every mixed team. Long DeepONets stopped at their wall-time caps; no verified loss plateau.
All columns compare the NN curve with the same IC Cantera curve. First graph row has only the ANN10k result.